Wasif Faisal

Computer Science | Payment Security | Threat Intelligence

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Profile

Computer science researcher focused on payment security and cyber threat intelligence. Investigates payment fraud and authentication weaknesses; turns evidence into practical risk recommendations. Seeking to protect digital payments, authentication pipelines, and customer transactions through payment security, fraud intelligence, authentication research, and technology-risk analysis.

Education

BRAC University 2026
B.Sc. in Computer Science Dhaka, Bangladesh
Completed all required B.Sc. credits in Summer 2026.
B.Sc. thesis: *CNP-SHIELD: Investigating Card-Not-Present (CNP) Vulnerabilities in Payment Gateways*; project lead and lead author.
Thesis supervisors: Md. Akhtaruzzaman Adnan, Senior Lecturer; S. M. Taiabul Haque, PhD, Professor.

Relevant Experience

Independent Payment Security Researcher (self-directed) Jan 2018–Present
Payment-fraud and cyber threat-intelligence research Remote

- Research publicly observable payment fraud, including card testing, attempts to identify valid card numbers, and misuse of stolen credentials.
- Review payment and authentication processes to check whether systems verify who is making each request and link it to the correct transaction; identify failure scenarios and signs of abuse.
- Translate findings into clear reports and practical recommendations for fraud, risk, product, and security teams, explaining how technical weaknesses create payment risk.

Research and Projects

The 3DS Oracle: BIN Attack via the Network 2026
Lead author; arXiv preprint. Examines how client-visible EMV 3-D Secure responses can reveal pre-authorization card and issuer-control signals; combines standards analysis, formal modeling, and bounded evaluation to develop payment-risk and mitigation recommendations.

Prefix Count Limits Can Increase First-Hit Discovery in Card Reissuance 2026
Lead author; arXiv preprint (arXiv:2609.31398; under review). Identified situations in which caps on how many cards can share a number prefix can make active cards easier to find in a BIN attack (attempts to identify valid card numbers); demonstrated the effect in simulations using 50,000 synthetic candidates. The findings help banks and other card issuers assess reissuance policies for card-number discovery risk, rather than relying on count limits alone.

EMV 3-D Secure Protocol Lab Public research and education resource
Built a public, vendor-neutral reference site for comparing 3-D Secure participants, messages, and decision points across published specification versions.

Leadership and Campus Activities

Assistant Director of Marketing | *BUNSC, BRAC University*
Senior Executive (Research & Publication) | *BUEDF, BRAC University*
Senior Executive (HR) | *BULDF, BRAC University*
Organizing Committee Volunteer | *BRAC University 17th Convocation*

Academic Service

Artifact Evaluation Program Committee Reviewer, ACM CCS 2026-B 2026
ACM Conference on Computer and Communications Security | *BRAC University*
Assess research software and datasets against paper claims; check that they work as documented and can be reproduced, while maintaining confidentiality.

Skills

Communication and risk: Coordinated vulnerability disclosure with CERT/CC, EMVCo, and vendors; communicated security findings across organizations; technical writing and evidence-based reporting; open-source threat intelligence, payment-fraud analysis, evidence assessment, technology-risk assessment, and control recommendations.
Data and tools: Python, SQL/SQLite, formal verification (ProVerif, Tamarin), Git, Linux, PowerShell, LaTeX, Burp Suite, and browser developer tools.
Office and productivity: Microsoft Word, Excel, PowerPoint; Google Docs, Sheets, and Slides.